Figure & Visual Practice

Many STAAR math questions include a **graph, diagram, or figure.** These items give you practice reading real visuals. (All figures below are exact, drawn-to-scale diagrams.) Answers are on the last page of this set.

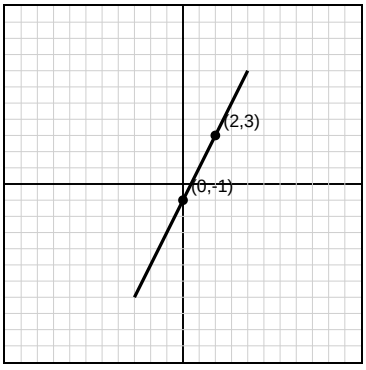
[IMAGE BOX — cover illustration]

Image prompt: "Clean, modern black-and-white book-cover graphic for a Texas STAAR Grade 8 study guide: a bold Texas star, an open book, and simple math/reading icons (ruler, pencil, speech bubble). Flat vector style, high contrast, no text."

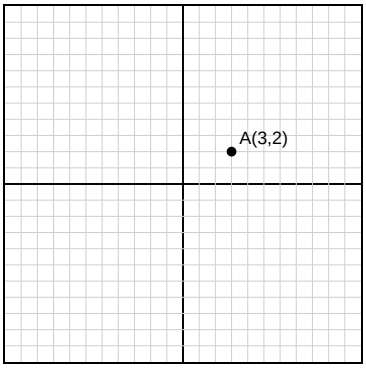
(Drop a real cover image here, or keep this prompt for a designer / image tool.)

Note: The box above shows how an illustration area is marked with an **image prompt** when a drawn diagram is not enough. The math figures in this book are exact diagrams, so they are drawn directly rather than generated.

Coordinate Grids

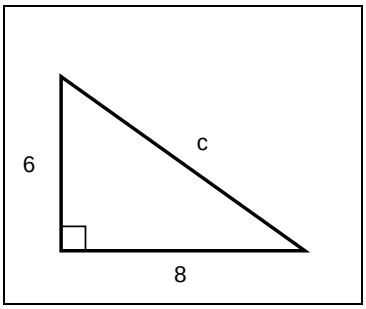


1. The line is graphed through the two marked points. What is its **slope**?
A. 1 B. 2 C. 3 D. 4

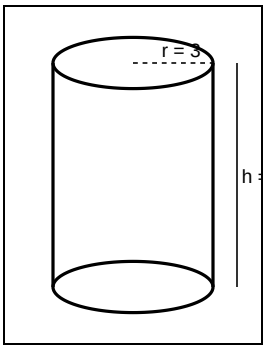


2. Point A is shown. What are the coordinates of its **reflection over the x-axis**? A. (−3, 2) B. (3, −2) C. (−3, −2) D. (2, 3)

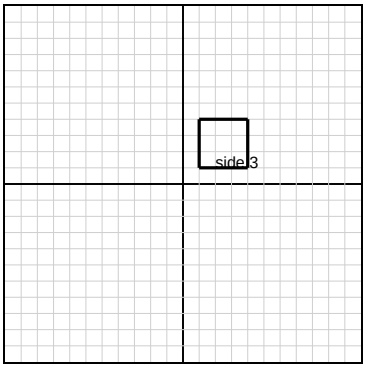
Geometry Figures



3. In the right triangle, the legs are 6 and 8. What is the length of the hypotenuse *c*? A. 10 B. 12 C. 14 D. 48

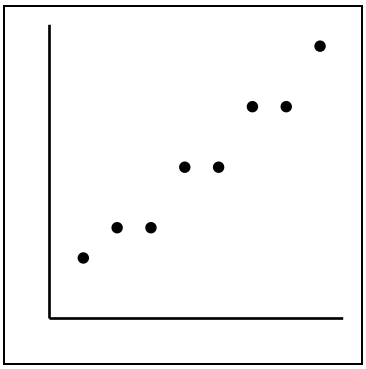


4. The cylinder has radius 3 and height 10. What is its **volume**? ($\pi \approx 3.14$)
A. 94.2 B. 188.4 C. 282.6 D. 565.2

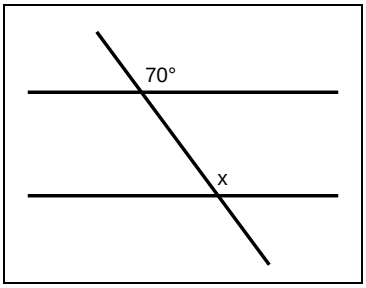


8. The square shown has a side of 3. If it is dilated by a scale factor of 2, what is the **area of the image**? A. 12 B. 18 C. 24 D. 36

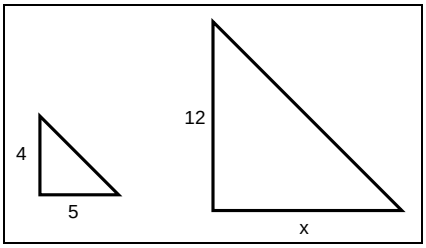
Data & Angles



5. The scatterplot shows the relationship between two variables. What **association** does it show? A. Positive linear B. Negative linear C. No association D. Non-linear



6. Two parallel lines are cut by a transversal. One angle measures 70°. What is the measure of angle *x* (its corresponding angle)? A. 20° B. 70° C. 110° D. 180°



7. The two triangles are similar. The small triangle has sides 4 and 5; on the large triangle the side matching the 4 is 12. What is the length *x* that matches the 5? A. 13 B. 14 C. 15 D. 20

Figure Practice — Answer Key

1. B *(8.4C ★)* — From (0,−1) to (2,3): rise 4, run 2 → slope = 4/2 = **2**.

2. B *(8.10C ★)* — Reflecting A(3,2) over the x-axis flips the y-sign → **(3, −2)**.

3. A *(8.7C ★)* — $c^2 = 6^2 + 8^2 = 36 + 64 = 100$ → c = **10**.

4. C *(8.7A ★)* — $V = \pi r^2 h = 3.14 \times 9 \times 10 =$ **282.6**.

5. A *(8.11A ○)* — The points rise left to right → **positive linear** association.

6. B *(8.8D ○)* — Corresponding angles formed by a transversal are equal → **70°**.

7. C *(8.3A ○)* — Scale factor = 12/4 = 3, so x = 5 × 3 = **15**.

8. D *(8.10D ○)* — Area scales by k^2: original area 3×3 = 9, image = $9 \times 2^2 =$ **36**.